«part def»

<LA-00> Logical Nanosatellite EPS Engineering Framework :> 'Specified Nanosatellite EPS Engineering Framework'

doc

Logical specialization of the complete framework black box. The parts below define the initial responsibility decomposition. Their interactions, ports, stores, properties, and state behavior are refined in the following logical architecture steps.

attributes

^declared minimum execution environment : String
^declared runtime dependencies : String
^<MOP-01> valid trace coverage : Percent
^<MOP-02> reference analysis reproduction success : Percent
^<MOP-03> open critical discrepancy count : Count
^<MOP-04> completed contrasting case count : Count
^<MOP-05> undeclared adaptation count : Count
^<MOP-06> correct change turnaround time : Hours
^<MOP-07> case adaptation effort : PersonHours
^<MOP-08> independent reproduction effort : PersonHours
^<MOP-09> framework adoption effort : PersonHours
^<MOP-10> direct framework cost : CAD
^<MOP-11> consistency assurance effectiveness : Percent

items

^requiredRecords : 'Controlled Framework Record Set'

parts

^systemModel : 'Versioned System Model'
^analysisTools : 'Analysis Tools'
^evidenceRepository : 'Evidence Repository'
missionApplicationInterface : 'Mission Application Interface Manager'
researchGovernanceInterface : 'Research Governance Interface Manager'
independentReviewInterface : 'Independent Review Interface Manager'
modelControlInterface : 'Model Control Interface Manager'
analysisExecutionInterface : 'Analysis Execution Interface Manager'
evidenceCustodyInterface : 'Evidence Custody Interface Manager'
evidenceChainManager : 'Evidence Chain Manager'
caseConfigurationManager : 'Case Configuration Manager'
analysisReproductionManager : 'Analysis Reproduction Manager'
designChangeAssessmentManager : 'Design Change Assessment Manager'
candidateEvaluationManager : 'Candidate Evaluation Manager'
designReviewManager : 'Design Review Manager'
workflowControlManager : 'Workflow Control Manager'
transferAssuranceManager : 'Transfer Assurance Manager'
baselineAndConfigurationManager : 'Baseline and Configuration Manager'
evidenceAndProvenanceManager : 'Evidence and Provenance Manager'
evaluationMeasurementManager : 'Evaluation Measurement Manager'

ports

^missionApplication : FrameworkBoundaryInterfaces::'Mission Application Port'
^researchGovernance : FrameworkBoundaryInterfaces::'Research Governance Port'
^independentReview : FrameworkBoundaryInterfaces::'Independent Review Port'
^modelControl : FrameworkBoundaryInterfaces::'Model Control Port'
^analysisExecution : FrameworkBoundaryInterfaces::'Analysis Execution Port'
^evidenceCustody : FrameworkBoundaryInterfaces::'Evidence Custody Port'

perform actions

^maintainEvidenceChain : FrameworkBlackBoxFunctions::'Maintain Evidence Chain'
^configureFrameworkForCase : FrameworkBlackBoxFunctions::'Configure Framework for EPS Case'
^reproduceDesignatedAnalysis : FrameworkBlackBoxFunctions::'Reproduce Designated Analysis'
^assessControlledDesignChange : FrameworkBlackBoxFunctions::'Assess Controlled Design Change'
^compareCandidateArchitectures : FrameworkBlackBoxFunctions::'Compare Candidate EPS Architectures'
^supportEarlyDesignReview : FrameworkBlackBoxFunctions::'Support Early Design Review'

exhibit states

^operationalControl : FrameworkOperationalState::'Framework Operational Control'